The Linear Antenna—Eighty Years of Progress

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Abstract—**Progress in understanding the basic properties of cylindrical antennas and arrays is reviewed from the time of Hertz to the present. The infinitesimal doublet, antennas with assumed sinusoidal currents, boundary-value solutions and recently developed approximations for isolated antennas, circular, curtain, and Yagi arrays are discussed and illustrated. It is concluded that useful quantitative methods for treating cylindrical antennas of finite cross section and arrays of such antennas are now available. These take account of actual distributions of current and the effects of mutual coupling on them, the driving-point admittances, and the field patterns.**

INTRODUCTION

A FULL CENTURY has elapsed since Maxwell [1] formulated his celebrated equations that continue to provide the foundations of classical electromagnetism. The wave phenomena deduced from them were first verified experimentally by Hertz [2] some 20 years later using a center-driven wire 60 cm long terminated at each end by a metal plate 40 cm square. The damped oscillations in this antenna were excited by a spark gap and had a fundamental frequency of 53.5 MHz or a wavelength of 5.6 meters. It is significant that there is still no complete analysis of the electrical properties of Hertz's dipole or, in fact, of many other geometrically comparable end-loaded radiators. During the 80 years since Hertz the properties of electromagnetic waves have been studied over a broad range of frequencies and under a great variety of conditions. Nevertheless, until quite recently relatively little interest was shown in the analytically complicated and experimentally difficult problems associated with the macroscopic sources which generate them.

It is the purpose of this paper to review critically certain basic aspects of the properties of the cylindrical antenna and arrays of such elements and to relate these to present and future work. However, no general survey of the advances in antenna design or technology is contemplated.

GENERAL FORMULATION OF ANTENNA THEORY

The primary purpose of an antenna is to provide suitably localized and oriented paths for oscillating electric currents. The distributions in amplitude and phase of accelerated charges moving along these paths determine not only the admittance, but also all of the characteristics of the electromagnetic field, including those associated with the directional properties of transmitters and receivers.

The field maintained by a distribution of axial currents in the cylindrical antenna shown in Fig. 1 is conveniently determined from Maxwell's equations with the intermediary of the vector potential $\boldsymbol{A}=\hat{z}A_z$ which satisfies the Helmholtz equation $(\nabla^2+\beta^2)A_z=0$ at all points outside the antenna. The wave number $\beta=\omega\sqrt{\mu\varepsilon}$ characterizes the infinite homogeneous medium in which the antenna is located. When the medium is air, μ and ε reduce to μ_0 and ε_0; when the medium is a conducting dielectric, ε is replaced by $\varepsilon-j\sigma/\omega$ where $\varepsilon=\varepsilon_0\varepsilon_r$ is the dielectric constant of the medium and σ is its conductivity. Let the cylindrical coordinates ρ, Φ, z be used; note that rotational symmetry with respect to Φ obtains. The vector potential at ρ, z due to the total axial current $I(z')$ in the perfectly conducting, thin-walled tubular antenna is given by

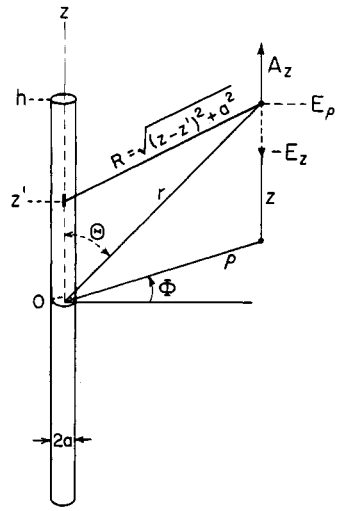

Fig. 1. Cylindrical antenna and coordinates.

$$A_z(\rho, z) = \frac{\mu}{4\pi}\int_{-h}^{h} I(z')K(z, z')dz' \tag{1}$$

where

$$K(z, z') = \frac{1}{2\pi}\int_0^{2\pi} \frac{e^{-j\beta R(\Phi')}}{R(\Phi')}\, d\Phi' \doteq \frac{e^{-j\beta R}}{R} \tag{2}$$

with

$$R(\Phi') = \sqrt{R^2 + a^2 - 2a\rho\cos\Phi'}$$
$$R = \sqrt{(z - z')^2 + \rho^2}. \tag{3}$$

The approximate form of the kernel on the right in (2) is usually adequate for thin tubular or solid antennas for all values of $\rho \geq a$. Once the vector potential is known, the components of the electromagnetic field can be obtained by differentiation. In cylindrical coordinates they are

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$$E_z(\rho, z) = \frac{j\omega}{\beta^2}\left(\frac{\partial^2}{\partial z^2} + \beta^2\right) A_z(\rho, z); \quad E_\rho(\rho, z) = \frac{\partial^2 A_z(\rho, z)}{\partial \rho \partial z};$$
$$B_\Phi(\rho, z) = \frac{-\partial A_z(\rho, z)}{\partial \rho}. \tag{4}$$

The corresponding components of the electric field in the spherical coordinates r, θ, Φ are

$$E_r(r, \theta) = E_z \cos\theta + E_\rho \sin\theta;$$
$$E_\theta(r, \theta) = -E_z \sin\theta + E_\rho \cos\theta. \tag{5}$$

The entire evaluation of the electromagnetic field clearly depends on the *prior* knowledge of the distribution of current $I(z)$ along the antenna. Moreover, if the antenna is driven at $z=0$ by a generator that maintains the EMF V_0^e, the driving-point admittance is $Y_0 = I(0)/V_0^e$, which also depends on a knowledge of the current. The same is true of the power in the load of a receiving antenna. It appears, therefore, that one of the most important quantities to be determined for an antenna is the distribution of current.

Hertz and the Oscillating Doublet

Hertz and the investigators who followed him for 40 years concerned themselves principally with the properties of electromagnetic waves at relatively large distances from the source. For this purpose, the structure of the antenna was unimportant so that it was conveniently replaced by a model, the oscillating doublet or Hertzian dipole, which consists of a short filament of current with a uniform amplitude oscillating at a frequency f between extensionless periodically varying concentrations of charge Q with opposite signs. That is, $I(z) = I(0) = j\omega Q(h)$ and $R \doteq r$, the distance from the center. It is readily shown that subject to the conditions, $a^2 \ll h^2 \leq r^2$,

$$A_z(r) \doteq \frac{\mu_0 h I(0)}{2\pi} \frac{e^{-j\beta r}}{r} = \frac{j\omega\mu_0 p_z}{2\pi} \frac{e^{-j\beta r}}{r} \tag{6}$$

where, in the expression on the right, the periodically varying current with the amplitude $I(0)$ in the short length h has been replaced by an equivalent doublet with electric dipole moment $p_z = 2hQ(h)$ so that $2hI(0) = j\omega p_z$. If h is allowed to approach zero while $Q(h)$ and $I(0)$ increase without limit to maintain the product $I(0)h$ at an assigned value, an ideal infinitesimal doublet is obtained. The electromagnetic field of such a doublet was determined by Hertz in 1888. In the spherical coordinates r, θ, Φ shown in Fig. 2, it is given by

$$E_r \doteq \frac{\omega\mu I(0)h}{2\pi\beta}\left(\frac{2}{r^2} - \frac{j2}{\beta r^3}\right) e^{-j\beta r} \cos\theta \tag{7a}$$

$$E_\theta \doteq \frac{j\omega\mu I(0)h}{2\pi\beta}\left(\frac{\beta}{r} - \frac{j}{r^2} - \frac{1}{\beta r^3}\right) e^{-j\beta r} \sin\theta \tag{7b}$$

$$B_\Phi \doteq \frac{j\mu I(0)h}{2\pi}\left(\frac{\beta}{r} - \frac{j}{r^2}\right) e^{-j\beta r} \sin\theta. \tag{7c}$$

Note that when applied to an antenna of finite size, however small, the following conditions must be met:

$$a^2 \ll h^2 \leq r^2. \tag{8}$$

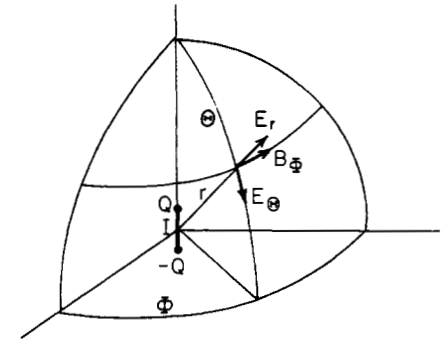

Fig. 2. Hertzian dipole and spherical components of electromagnetic field.

The terms in (7a)–(7c) with $1/r$ as a factor are known as the radiation or far-zone field; all other terms constitute the induction or near-zone field. Note, however, that the $1/r^3$ terms occur only in the field of an electrically short dipole; they do not appear in the near-zone field of longer antennas. If less severe conditions than (8) had been imposed, still higher powers of $1/r$ would have occurred.

The real, time-average power P_r transferred from a transmitting system (consisting of an antenna and generator with an arbitrary closed surface Σ) to the universe outside Σ is given by the Poynting-vector theorem [3],

$$P_r = \mathrm{Re} \int_\Sigma \hat{n} \cdot \boldsymbol{S}\, d\Sigma \tag{9}$$

where the complex Poynting vector $S = \frac{1}{2}\mu_0(\boldsymbol{E} \times \boldsymbol{B}^*)$. If (9) is applied to the Hertzian dipole in air, with Σ chosen to be a great spherical shell in the far zone, the integration is easily carried out. The result is $P_r = \frac{1}{2}I^2(0)R_0^e$ where $R_0^e = 20\beta^2 h^2$ ohms is known as the radiation resistance of the doublet. Actually, any surface Σ that completely encloses the doublet could be chosen and the result would have to be the same. However, if a smaller surface near the doublet is used, the integration is much more complicated and if the surface is too small the conditions (8) may be violated so that no accurate expressions for the field are available. If the medium is dissipative, the integral (9) still gives the power transferred across Σ, but it now depends on the shape and size of Σ, owing to dissipation in the part of the medium within Σ.

Ever since the days of Hertz, the infinitesimal doublet has been substituted for analytically more difficult but physically realizable antennas in studies that involve sources in various media. Recent examples include the antenna near the interface of a dissipative medium and air [4] and many investigations relating to antennas in plasmas. It must not be overlooked that this procedure avoids major aspects of the antenna problem; it does not solve them.

Finite Antennas with Assumed Currents

In the late 1920's a new phase began for antenna theory when the importance of the finite lengths of radiating elements was recognized. However, no attempt was made to determine the actual distributions of current analytically. Instead, a convenient sinusoidal distribution was assumed,

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based in part on measured values, in part on its adequacy in special cases, and in part on the mistaken notion that when a section at the open end of a two-wire line is bent outward to form a center-driven dipole, there is no significant change in the current. With an assumed current in the form

$$I(z) = I_m \sin \beta(h - |z|) \tag{10}$$

the integral (1) is readily expressed in terms of sine and cosine integral functions. More significantly, with (10) substituted in (1), the complete electromagnetic field can be evaluated in closed form by the application of (4) directly to the integrals. The resulting expressions were first obtained for the half-wave dipole by Carter [5] in 1932, and for an antenna of arbitrary length by Brown [6] in 1937. They are

$$E_\rho = \frac{jI_m\zeta}{4\pi\rho}\left(\frac{z-h}{R_{1h}}e^{-j\beta R_{1h}} + \frac{z+h}{R_{2h}}\cdot e^{-j\beta R_{2h}} - \frac{2z}{R_0}\cos\beta h e^{-j\beta R_0}\right) \tag{11a}$$

$$E_z = \frac{-jI_m\zeta}{4\pi}\left(\frac{e^{-j\beta R_{1h}}}{R_{1h}} + \frac{e^{-j\beta R_{2h}}}{R_{2h}} - 2\cos\beta h\frac{e^{-j\beta R_0}}{R_0}\right) \tag{11b}$$

$$B_\Phi = \frac{jI_m\mu_0}{4\pi\rho}\left(e^{-j\beta R_{1h}} + e^{-j\beta R_{2h}} - 2\cos\beta h e^{-j\beta R_0}\right) \tag{11c}$$

where $R_{1h} = \sqrt{(z-h)^2+\rho^2}$, $R_{2h} = \sqrt{(z+h)^2+\rho^2}$, and $R_0 = \sqrt{z^2+\rho^2}$. Note that the only distances that occur in (11a)–(11c) are from the point of observation to the ends and the center of the antenna. No approximations are involved in these expressions other than those implicit in the assumed current (10). For an electrically short antenna with $\beta h \ll 1$, they lead to (7a)–(7c) with 4π in the denominators instead of 2π and subject to the additional conditions (8). When (11a)–(11c) are transformed from cylindrical to spheroidal coordinates a physically attractive picture of expanding spheroidal wave fronts is obtained. This was first suggested by Riazin [7] in 1937 and further developed by King [8], [9].

It is readily verified from (11b) that E_z has a maximum value at $\rho = 0$, a value which decreases only very slowly as ρ is increased. It follows that the boundary condition $E_z = 0$ on the surface $\rho = a$ of a perfectly conducting cylindrical antenna cannot be satisfied by (11b). Clearly, the field (11a)–(11c) determined by Carter and Brown is not the field of a center-driven, perfectly conducting cylinder of radius a. Actually, (11a)–(11c) is the correct field of an antenna with a continuous distribution of EMF's along its length appropriate to maintain the required sinusoidal current. For such a structure, which is simultaneously an antenna and a distributed generator, a radiation resistance is readily defined with (9). Since the expressions for the electromagnetic field are exact (for the postulated sinusoidally distributed current), the surface Σ can be chosen to be a cylinder of small radius a enclosing the antenna instead of the surface of a great sphere. If the small circular top and bottom surfaces are neglected, and use is made of the relation $I_z = 2\pi a\mu_0^{-1}(B_\Phi)_{\rho=a}$—which implies that the cylinder, $-h \le z \le h$, $\rho \le a$, is highly conducting—(9) may be expressed as follows:

$$P = P_r + jP_i = -\frac{1}{2}\int_{-h}^{h}(E_z)_{\rho=a}I_z^*\,dz = \frac{1}{2}|I_m|^2 Z_m^e = \frac{1}{2}|I(0)|^2 Z_0^e \tag{12}$$

where I_m is the amplitude of the sinusoidal current and $Z_m^e = R_m^e + jX_m^e$ is the radiation impedance of the antenna referred to this maximum. Note that only the real part of (12) actually corresponds to (9). Alternatively, since from (12) $I(0) = I_m \sin \beta h$, $Z_0^e = Z_m^e \csc \beta h$ is the radiation impedance referred to the current at the center ($z = 0$) of the antenna. In the absence of a single localized generator with a definite pair of terminals, it is strictly not possible to define a driving-point impedance. Nevertheless, $Z_0^e = R_0^e + jX_0^e$ as defined in (12) is often treated as if it were the driving-point impedance of an antenna with a single generator at its center. Evidently, the extent to which this is a meaningful substitution depends entirely on the degree in which the assumed distribution of current in (12) approximates the actual current in a center-driven antenna. It turns out that for determining certain properties of very thin antennas over a very limited range of lengths (12) is an acceptable engineering approximation. The formulation (12) is known as the EMF method.

The Thin Cylindrical Antenna as a Boundary-Value Problem

The correct method for determining the properties of a center-driven cylindrical antenna is to determine the current which generates the electromagnetic field that actually satisfies the boundary condition $E_z = 0$ at $\rho = a$, $-h \le z \le h$. The general expression for E_z in cylindrical coordinates is given in (4) in terms of the vector potential. If, in order to have a mathematically tractable problem, it is assumed that the perfectly conducting cylinder is center driven by the discontinuity in scalar potential,

$$V_0^e = \lim_{z\to 0}[\phi(z) - \phi(-z)],$$

the boundary condition becomes

$$E_z(a, z) = -V_0^e\delta(z) = \frac{j\omega}{\beta^2}\left(\frac{\partial^2}{\partial z^2} + \beta^2\right)A(a, z); \quad |z| \le h. \tag{13}$$

A solution of this equation can be equated to the general integral (1) evaluated at $\rho = a$ to obtain the following integral equation for the current $I(z)$:

$$4\pi\mu_0^{-1}A_z(a, z) = \frac{j4\pi}{\zeta_0}\left[C_1\cos\beta z + \frac{1}{2}V_0\sin\beta|z|\right] = \int_{-h}^{h} I(z')K(z, z')\,dz'. \tag{14}$$

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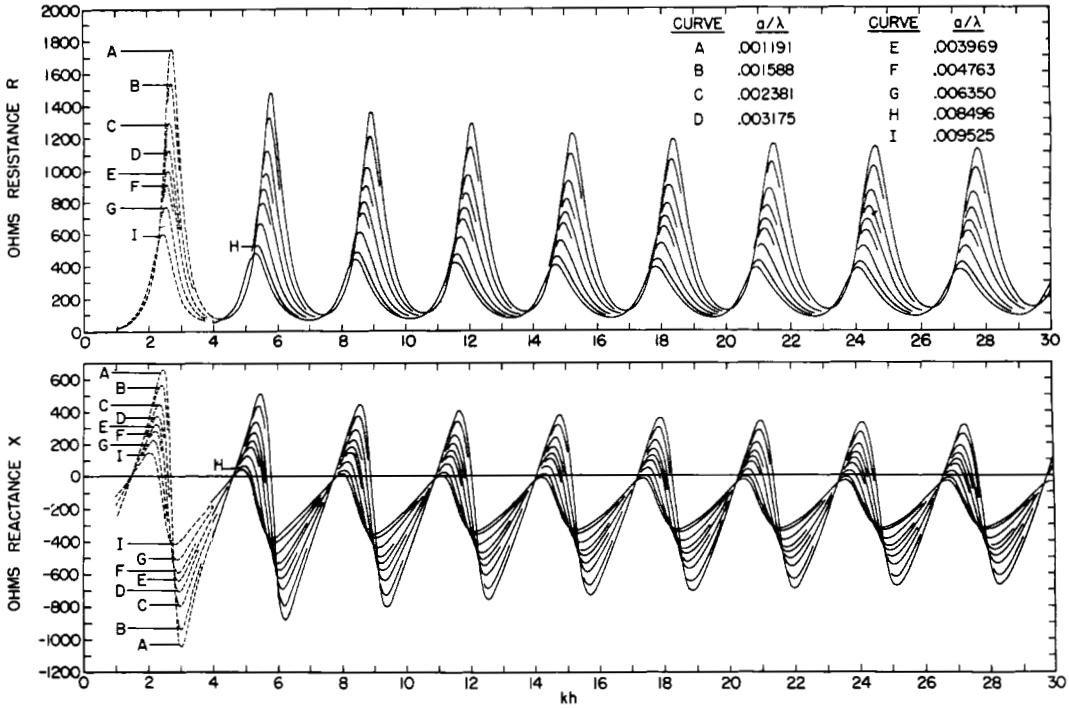

Fig. 3. Resistance and reactance of cylindrical antenna with half-length h and radius a. Dotted line: King-Middleton second-order theory. Solid line: Wu theory.

The symmetry conditions $I(-z)=I(z)$, $A_z(-z)=A_z(z)$ have been invoked in (14). The constant C_1 is to be evaluated from the condition $I(\pm h)=0$.

Approximate solutions of (14) have been obtained by many methods beginning in 1937 with the work of L. V. King [10] and in 1938 with that of Hallén [11] and continuing to the present. Among the various analytical procedures that have been used for electrically thin antennas ($a \ll h$, $\beta a \ll 1$) are three methods that permit successive improvement. These are iteration (Hallén [11], King and Middleton [12]), Fourier series (Duncan and Hinchey [13]), and numerical integration (Mei [14]). An increase in the number of iterations, terms in the series, or points along the antenna where the equation is satisfied, exactly improves the accuracy of the solution if the procedures converge. Since none of these methods was convenient to handle electrically long antennas ($\beta h \geq 2\pi$), a solution based on the Wiener-Hopf technique (Wu [15]) proved to be a valuable extension. Variational methods (Storer [16], Tai [17]), which provide a means of improving an already known result, yielded useful supplementary confirmation of the solutions obtained by other procedures. Representative graphs of the impedances obtained by iteration (second-order King-Middleton [12]) for shorter lengths, and by the Wiener-Hopf technique (Wu [15]) for longer antennas in air, are shown in Fig. 3. Corresponding data for antennas in dissipative media have been reported [18]. Results obtained by Fourier series expansion, direct numerical integration [14], and the variational method [16], [17] are in good agreement.

In order to relate theoretical curves such as those in Fig. 3 to measurements made on actual antennas driven from transmission lines, an extensive and detailed study of the special properties of both theoretical and experimental models was required. In order to illustrate the problems involved, consider the iterative solution of (14) in the following form given by King and Middleton [12]:

$$I(z)=\frac{j2\pi V_0^e}{\zeta_0 \Psi_k}\left[\frac{\sin\beta(h-|z|)+M_1(z)/\Psi_k+M_2(z)/\Psi_k^2+\cdots}{\cos\beta h+A_1(h)/\Psi_k+A_2(h)/\Psi_k^2+\cdots}\right] \quad (15)$$

where Ψ_k is a real parameter, the $M(z)$ are iterated integrals that are functions of z, and the $A(h)$ are iterated integrals that are functions of h. Detailed formulas are in the literature [12]. In the original evaluation to obtain the curves of $Z=V_0^e/I(0)$ shown in Fig. 3, the approximate kernel in (2) was used with two iterations. More recently [19], the admittance $Y=1/Z=G+jB$ was evaluated from (15) with up to thirty iterations carried out by a high-speed computer. Both the approximate and the exact kernels in (2) were used. The results showed that when $\beta a \leq 0.02$, G converged to a satisfactory final value with either kernel in two or at most three iterations, but that B diverged with each iteration adding an amount proportional to the circumference of the antenna. Since the theoretical model used in all the theoretical analyses included an idealized delta-function generator that is equivalent to a discontinuity in scalar potential in parallel with a knife-edge capacitance $C_g=\infty$ at $z=0$, this is not surprising. The calculated Nth-order value of the susceptance includes $[B]_N$ for the antenna in parallel with $[B_g]_N$

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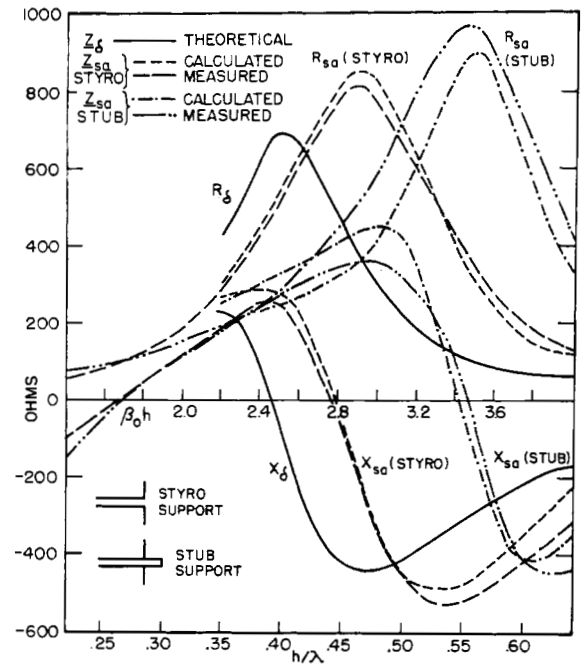

Fig. 4. Ideal theoretical impedance Z_δ and corrected apparent impedances Z_{sa} for two types of support for the antenna. Measured impedances are also shown (P. Conley [21], pp. 207 and 215–219).

for the generator. Since $[B]_N$ should converge approximately as rapidly as $[G]_N$, i.e., with $N=2$ or 3, whereas $[B_g]_N = [\omega C_g]_N \doteq [\omega 2\pi a C_g]_N$ approaches infinity as N is increased without limit, $[B]_N + [B_g]_N$ must diverge. In order to determine $[B]_N$, some means must be found to subtract $[B_g]_N$. A useful method is to obtain one accurate value of B for each thickness of the antenna experimentally at a convenient value of βh (e.g., the value yielding the first antiresonance) and then to correct all theoretical values of $[B]_N$ by this amount. Tables of Y and other properties of the antenna obtained in this manner are in the literature [19] for $N=2$.

The determination of correct experimental values of the admittance of a cylindrical antenna is not at all simple. Strictly, an admittance independent of the geometry of the junction of the antenna and the feeding transmission line does not exist. The adjacent parts of the antenna and the line are always coupled, and end effects exist on any line that does not extend to infinity. As a consequence, the admittance apparently terminating a transmission line as obtained by conventional methods is not the ideal independent admittance of the antenna. This latter may be defined in the hypothetical limit of "zero" line spacing by an extrapolation of measurements made with successively smaller distances between the conductors of the line. The actual limit is, of course, unattainable since the characteristic impedance of a transmission line vanishes at zero line spacing so that in the limit the normalized impedance of any finite load is infinite. Extensive theoretical and experimental studies of terminal-zone corrections for transmission lines of different types have been made and excellent agreement between corrected theoretical and experimental impedances has been obtained [20]–[22] for electrically thin antennas of any length [12], [23]. Examples of the apparent impedance Z_{sa} of a cylindrical antenna when center-driven from a two-wire line with and without a quarter-wave stub support are given in Fig. 4. The corrected theoretical curves obtained from the ideal theoretical impedance Z_δ with appropriate terminal-zone corrections are also shown.

An Engineering Approximation for Thin Cylindrical Antennas

Although the various solutions of the integral equation (14) yield quantitatively useful impedances and currents, the formulas for the latter are too complicated for use in the evaluation of electromagnetic fields and the methods themselves are too cumbersome to permit their ready extension to coupled antennas. What was needed was an expression for the current that would combine a measure of the simplicity of the sinusoidal current (10) with acceptable quantitative accuracy over a useful range of lengths and thicknesses of cylindrical antennas. Such a solution was obtained by King [24] in 1959 and further improved [25] in 1966. It proceeds from a rearrangement of the integral equation (14) in the form

$$\int_{-h}^{h} I(z')K_d(z, z')dz' = \frac{j4\pi}{\zeta_0 \cos \beta h}\left[\frac{1}{2} V_0 \sin \beta(h - |z|) + U(\cos \beta z - \cos \beta h)\right] \quad (16)$$

where

$$K_d(z, z') = K(z, z') - K(h, z') \quad (17)$$

and

$$U = \frac{-j\zeta_0}{4\pi}\int_{-h}^{h} I_z(z')K(h, z')dz'. \quad (18)$$

From a careful study of the properties of the real and imaginary parts of the difference kernel $K_d(z, z')$ and of the behavior of the associated parts of the integral in (16), it was shown that a very good approximate solution for $\beta h \leq 5\pi/4$ is

$$I(z) \doteq \frac{j2\pi V_0}{\zeta_0 \Psi_{dR} \cos \beta h}\{\sin \beta(h - |z|) + T_U(\cos \beta z - \cos \beta h) + T_D(\cos \tfrac{1}{2}\beta z - \cos \tfrac{1}{2}\beta h)\} \quad (19)$$

where Ψ_{dR} is a real constant, and T_U and T_D are complex constants [25]. The accuracy of this simple formula in representing the components of current in phase and in phase quadrature with the driving voltage is shown in Fig. 5 for a cylindrical antenna a full wavelength long. The agreement is seen to be excellent except for $I'(z)$ near $z=0$ and, hence, also for the susceptance B. The error may be corrected with a lumped susceptance as shown in Fig. 6 for the even

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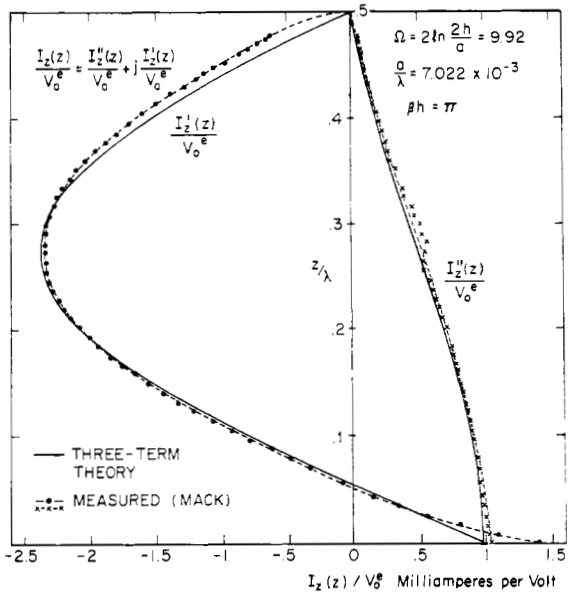

Fig. 5. Current distribution along a cylindrical antenna a full wavelength long. $I_{z_*}'(z)$ is in phase quadrature, $I_z''(z)$ in phase with the driving voltage V_0^e. Measurements by Mack [43].

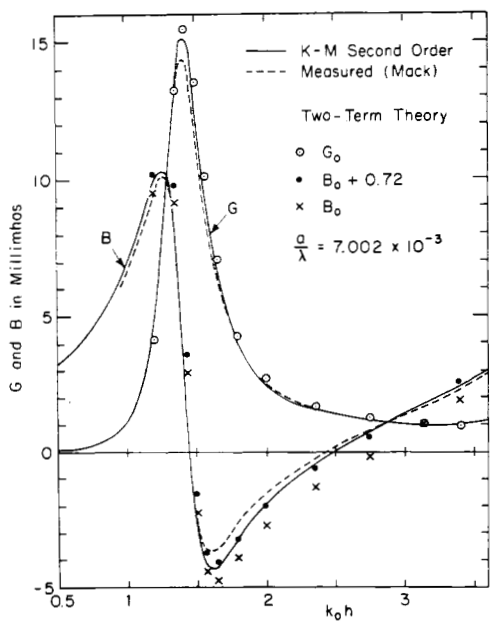

Fig. 6. King-Middleton second-order admittance, $Y = G + jB$, two-term admittance, $Y_0 = G_0 + jB_0$, and measured admittance of cylindrical antenna. Measurements by Mack [43].

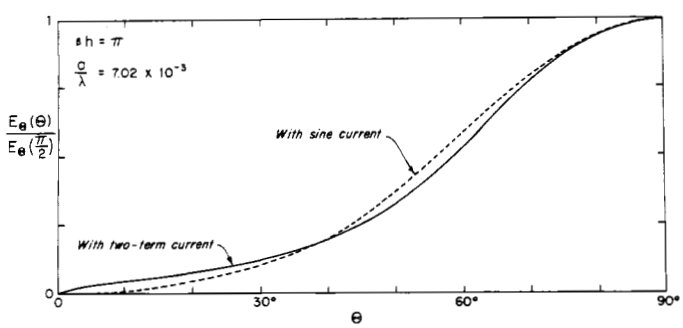

Fig. 7. Normalized magnitude of the far field of a full-wave antenna. Solid line: two-term theory. Dotted line: with sinusoidal current.

simpler and usually adequate two-term form obtained from (19) with $T_D = 0$. The method has been extended to asymmetrically driven [26] and resistive [27] antennas, and to antennas in dissipative media [28]. The underlying principle has also been applied to solve the integral equation for the tangential electric field near the surface of the perfectly conducting antenna [29].

The electromagnetic field can be evaluated directly from (19), both for points near the antenna [25], and in closed form for the far zone. For the latter, the two-term current (given by (19) with $T_D = 0$) is usually adequate. Referred to the current at $z = 0$, it is

$$E_\theta^r = \frac{j\zeta_0 I(0)}{2\pi} \frac{e^{-j\beta R_0}}{R_0} \left[\frac{F_m(\theta, \beta h) + T_U G_m(\theta, \beta h)}{\sin \beta h + T_U(1 - \cos \beta h)}\right], \quad (20)$$

where

$$F_m(\theta, \beta h) = \frac{\cos(\beta h \cos\theta) - \cos \beta h}{\sin \theta} \quad (21a)$$

$$G_m(\theta, \beta h) = \frac{\sin \beta h \cos(\beta h \cos\theta) \cos\theta - \cosh \beta h \sin(\beta h \cos\theta)}{\sin\theta \cos\theta}. \quad (21b)$$

The normalized magnitudes of the far-zone field as obtained from (20) and from the sinusoidal current (10) for a full-wave antenna are shown in Fig. 7.

Arrays of Dipoles with Identically Distributed Currents

Since the 1920's an extensive theory has been developed for the far-zone fields of arrays of antennas in which the distributions of current along all elements are alike and the phase of the current at each point along each element is the same as at the driving-point [30]–[34]. When these conditions are fulfilled, the individual field factors of all elements are identical and independent of the relative locations of the elements in the array. It is then possible to express the far field of the array as the simple product of the field factor $F(\theta, \beta h)$ of each and every element and an array factor $A(\theta, \Phi)$. Moreover, the mutual impedance between any pair of elements is independent of the presence of all other elements (since the assumed distributions of current are by definition unaffected by their presence). This greatly simplifies the analysis of the circuit properties of arrays [35]–[40]. In most cases, these basic postulates (which underlie essentially all of the many developments in the conventional theory of arrays) were implemented by assuming sinusoidal distributions of current along all elements. Evidently, the validity of conventional array theory depends crucially on the degree in which this postulate is fulfilled in actual arrays of antennas. It is, of course, exactly satisfied when the elements are identical infinitesimal doublets or when suitably distributed EMF's are provided along elements of finite length to maintain the required sinusoidal distribution of current in all environments.

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When the elements of an array are finite in length and cross section and are either center-driven (base driven over a ground plane) or parasitic, the distributions of current are not, in general, alike, nor is the phase constant along each antenna. As a consequence, the conventional formulas for the far field and for mutual and self-impedances lose their validity except under special circumstances. Specifically, when the elements are very thin and all are either resonant or electrically short, the errors remain tolerable.

A New Theory of Arrays of Cylindrical Dipoles with Finite Cross Sections

In order to provide a more satisfactory theory of arrays of N parallel identical elements, it is necessary to determine the distributions of current along all elements from boundary conditions like $E_{kz}(a, z) = V_{k0}^{e}\delta(z)$ on the surface of element k, $k=1, 2 \cdots N$. This leads to the N simultaneous integral equations

$$\int_{-h}^{h} \sum_{i=1}^{N} I_{iz}(z_i')K(z_k, z_i')dz_i' = \frac{-j4\pi}{\zeta_0}\left[C_k \cos \beta z_k + \frac{1}{2} V_{k0} \sin \beta|z_k|\right] \quad (22)$$

where $K(z_k, z_i')$ is given by (2) with z replaced by z_k, z' by z_i', and ρ by b_{ki}, the distance between the parallel axes of elements k and i. Note that $b_{kk}=a$, the radius of each element. These equations are difficult to solve in general. When the elements are arranged at the vertexes of a regular polygon, the N simultaneous integral equations can be replaced by N independent integral equations for N phase sequences [41], [42]. The solution of these equations leads to expressions for the currents in the arbitrarily driven elements of a circular array that are sums of N terms like (15) with suitable coefficients. These results are too complicated for practical use; moreover, they are restricted to circular arrays.

Fortunately, a simple extension of the two- and three-term approximations [(19) with and without $T_D=0$] has been developed for arrays of parallel elements [24]. It proceeds from a rearrangement of the integral equations (22) into the form

$$\int_{-h}^{h} \sum_{i=1}^{N} I_{iz}(z_i')K_d(z_k, z_i')dz_i' = \frac{j4\pi}{\zeta_0 \cos \beta h}\left[\frac{1}{2} V_{k0} \sin \beta(h - |z_k|) + U_k(\cos \beta z - \cos \beta h)\right] \quad k = 1, 2, \cdots N \quad (23)$$

where $K_d(z_k, z_i')$ is given by (17) with z replaced by z_k, z' by z_i' and a by b_{ki}, and

$$U_k = \sum_{i=1}^{N} U_{ki} = -j\frac{\zeta_0}{4\pi}\int_{-h}^{h} \sum_{i=1}^{N} I_{zi}(z_i')K(h_k, z_i')dz_i'. \quad (24)$$

When applied to the circular array of N elements located at the vertexes of an N-sided equilateral polygon, the N simultaneous equations become N independent equations in the N phase sequences. Each of these is like (16) with a kernel that consists of a sum of N terms. A study of the properties of the $N-1$ integrals in addition to the one in (16) reveals that, except for very closely spaced elements ($\beta b_{ik}<1$), all contributions to the current by mutual coupling are very well approximated by the trigonometric functions ($\cos \beta z - \cos \beta h$) and ($\cos \frac{1}{2}\beta z - \cos \frac{1}{2}\beta h$). This means that the current distribution in each of the N elements in each phase sequence is like (19), but with the complex coefficients T_U and T_D modified by its interaction with all of the other elements. It is now easily seen that a superposition of the currents in the N phase sequences to obtain the currents for an arbitrary circular array must again yield currents in all elements of the form (19), but with different sets of coefficients for each element. However, since the interaction between the N currents in an arbitrarily driven circular array includes a wide range of distances and relative positions, it is evident that the currents in the elements of any parallel array, including especially curtain arrays, must have distributions that are well approximated by

$$I_k(z) = A_k \sin \beta(h - |z|) + B_k(\cos \beta z - \cos \beta h) + C_k(\cos \tfrac{1}{2}\beta z - \cos \tfrac{1}{2}\beta h), \quad (25)$$

if the coefficients A_k, B_k, and C_k are determined to satisfy the integral equation specifically for element k. It turns out that these coefficients are readily evaluated and that they take account of the coupling between elements in a manner that is sufficiently general to represent correctly the often large differences in the distributions of current along elements that are not similarly situated in an array. For most purposes, a two-term representation with $C_k=0$ in (25) is quite adequate.

Applications to Practical Arrays

As an illustration of the accuracy of the two-term theory in the analysis of arrays, consider the self- and mutual impedances of a five-element circular array shown in Fig. 8. Theoretical values agree well with measurements made by Mack [43] for an array consisting of one driven and four parasitic elements. The graphs for an array of full-wave elements were chosen deliberately, since conventional sinusoidal theory is completely powerless to treat elements of this length. The associated field patterns in the equatorial plane are shown in Fig. 9 for the five-element parasitic array and for a two-element array. The agreement between theory and experiment is consistently good.

In Fig. 10 are shown the theoretically determined distributions of current along the eight full-wave elements of an endfire array [44]. The currents at the centers of the elements were required to be equal in magnitude and to increase progressively in phase. Note the difference in the distribution of current in the first and the eighth elements. Evi-

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dently a proportionately much larger fraction of the power is supplied to and radiated from the former.

The upper part of Fig. 11 shows graphs of the computed driving-point impedances of full-wave endfire arrays of 5, 9, 15, and 25 elements [44]. Since the driving-point currents are kept equal in magnitude, it is clear that a large fraction of the power to the entire array is actually supplied to element no. 1 since its resistance is many times that of any other, whereas the driving-point currents are all equal in magnitude. At the bottom of Fig. 11 are the impedances of the elements of the same array when the beam is turned to produce a broadside pattern. They are significantly different. Although the examples chosen have involved full-wave elements, the new theory is equally applicable to arrays of the more usual half-wave elements. In Fig. 12 are graphs of the impedances of the same endfire array, but made of half-wave instead of full-wave elements. Note the entirely different behavior. The forward element now has relatively the greatest amount of power, not the rear one as with the full-wave elements. Graphs of the field patterns for endfire arrays of 25 elements when calculated from the correctly represented different currents are shown in Fig. 13 for elements with three different lengths. The "ideal" patterns for arrays with assumed identical distributions of current are also shown. They are seen to differ only slightly in having sharp nulls instead of rounded minimums when $\beta h = \pi/2$, but very drastically when $\beta h = \pi$. In this case the minor lobe structure becomes essentially a small ripple on an almost uniform magnitude. Note that a considerably better approximation of the ideal pattern is obtained with full-wave elements when the driving voltages instead of the currents are assigned equal amplitudes and a progressive phase shift. This is a consequence of the fact that the coefficients A_k in (25) are determined entirely by the driving voltage and not at all by mutual interaction. It follows that the sinusoidal components of current (which are zero at the driving point but large out along the antennas) are specified instead of the shifted-cosine components (which are large at the driving point and small out along the antennas). Note that the field pattern is determined primarily by large currents out along the antenna.

A further test of the two-term theory was made with an array of 19 parasitic directors and one driven antenna arranged in the manner of a Yagi array. The amplitudes of the currents at the centers of the elements are shown in Fig. 14 for different lengths of the antennas when all were spaced at intervals of 0.2λ. Theoretical results are compared with measurements by Mailloux [45]. For the shorter lengths $h/\lambda = 0.16$ and 0.18 the agreement is excellent. When $h/\lambda = 0.20$ the theoretical and experimental points lie on curves that are similar in shape but are somewhat separated. However, if the theoretical points are calculated for $h/\lambda = 0.204$, an increase in length by less than the radius of the wire, almost perfect agreement is obtained. Comparable agreement is also obtained for the phases. The field patterns of the array are shown in Fig. 15. Because of the finite but large size of the ground plane and the great length of the array, Mailloux was unable to measure the field pattern in the far zone of the array. However, measurements were made in the far zone of each element. The two theoretical curves are for a true far-zone field and for a field calculated from far-zone formulas for each element. The agreement, even to the details of the minor lobe structure, is seen to be excellent—a necessary consequence of the correct representation of the individual currents.

For all parasitic elements in an array, the coefficient A_k in (25) vanishes since it is directly proportional to V_{k0}. This means that if account is to be taken of phase changes of the currents along such elements, both the coefficients B_k and C_k are required. An analysis of the Yagi-Uda array shown in Fig. 16 has been made with the three-term theory [46]. The array consists of a single parasitic reflector (no. 1) slightly longer than the half-wave driven element (no. 2) and a number of directors all of equal length and spacing. By varying the spacing b and the length $2h$ of the directors, computer printouts were obtained for such quantities as the front-to-back ratio, the forward gain, the input impedance, the field pattern, and the distributions of current on all elements. Figure 17 shows a set of contours drawn on the computer printout for the forward gain of a four-element array. The contours were drawn along paths of constant forward gain. A series of maximums and minimums occur at lengths near $h/\lambda = 0.43\lambda$ as the distance between the directors is increased. The forward gain for arrays with 1, 2, 4, and 8 directors when the latter have the length $h/\lambda = 0.43$ is shown in Fig. 18 as a function of the spacing b/λ. The distributions of current along the ten elements of a typical Yagi array are shown in Fig. 19. Each current is represented by the components in phase and in phase quadrature with the voltage V_{20} driving element 2. For the particular choice of parameters the amplitudes of the currents in the directors are seen to be substantially alike. Moreover, the phase distribution along each element turns out to be practically constant and the phase from element to element is almost that of a wave traveling along the array with the velocity of light. This is shown in Fig. 20. It is this behavior of suitably designed Yagi arrays that makes the application of surface-wave theory possible [45], [47], [48]. The field pattern of the array of ten elements is shown in Fig. 21.

These several examples serve to illustrate the power of the new approach to array theory. With the aid of a high-speed computer all significant properties of arbitrarily driven circular, curtain, and Yagi arrays can be determined with quantitative accuracy if the elements are no longer than $2h = 5\lambda/4$ and $\Omega = 2 \ln (2h/a) \geq 8$. This permits a 5-to-1 frequency ratio with antenna half-lengths in the useful range $(\lambda/8) \leq h \leq (5\lambda/8)$, and with vertical field patterns that have the principal lobe in the equatorial plane. The theory has been applied to coupled antennas in dissipative media [49]. Its extension to log-periodic arrays and planar arrays that include collinear and staggered elements is in progress. Note that the latter are complicated by currents that do not satisfy the symmetry condition $I(-z) = I(z)$ with respect to the center of each element.

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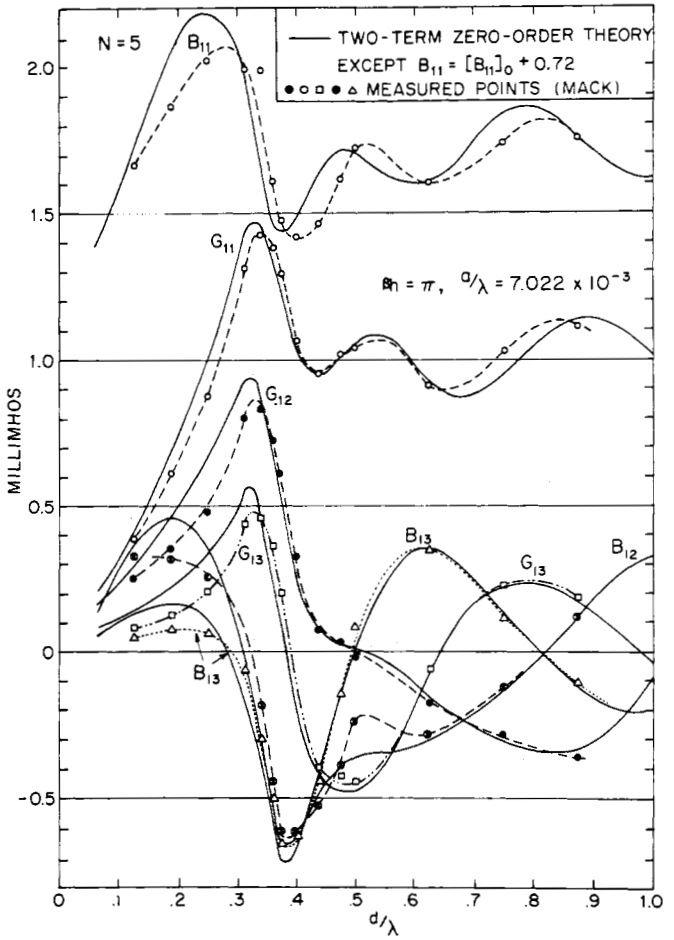

Fig. 8. Self- and mutual admittances of a five-element circular array as determined by the two-term theory and by measurement. d is the distance between adjacent elements. Measurements by Mack [43].

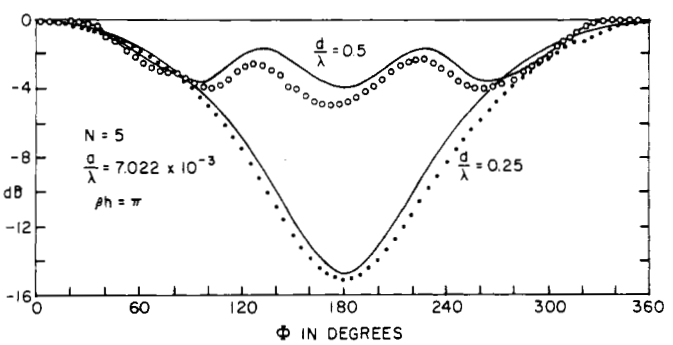

Fig. 9. Horizontal field pattern of parasitic arrays of N elements in a circle; d is the distance between adjacent elements. Solid line: two-term theory; experimental points from data by Mack [43].

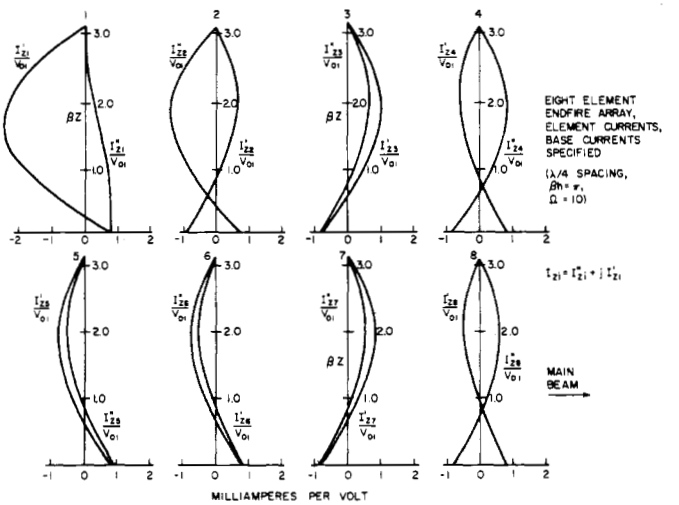

Fig. 10. Current distributions along an eight-element endfire array of full-wave antennas with currents at the driving point specified to be equal in amplitude and vary progressively in phase. Two-term theory.

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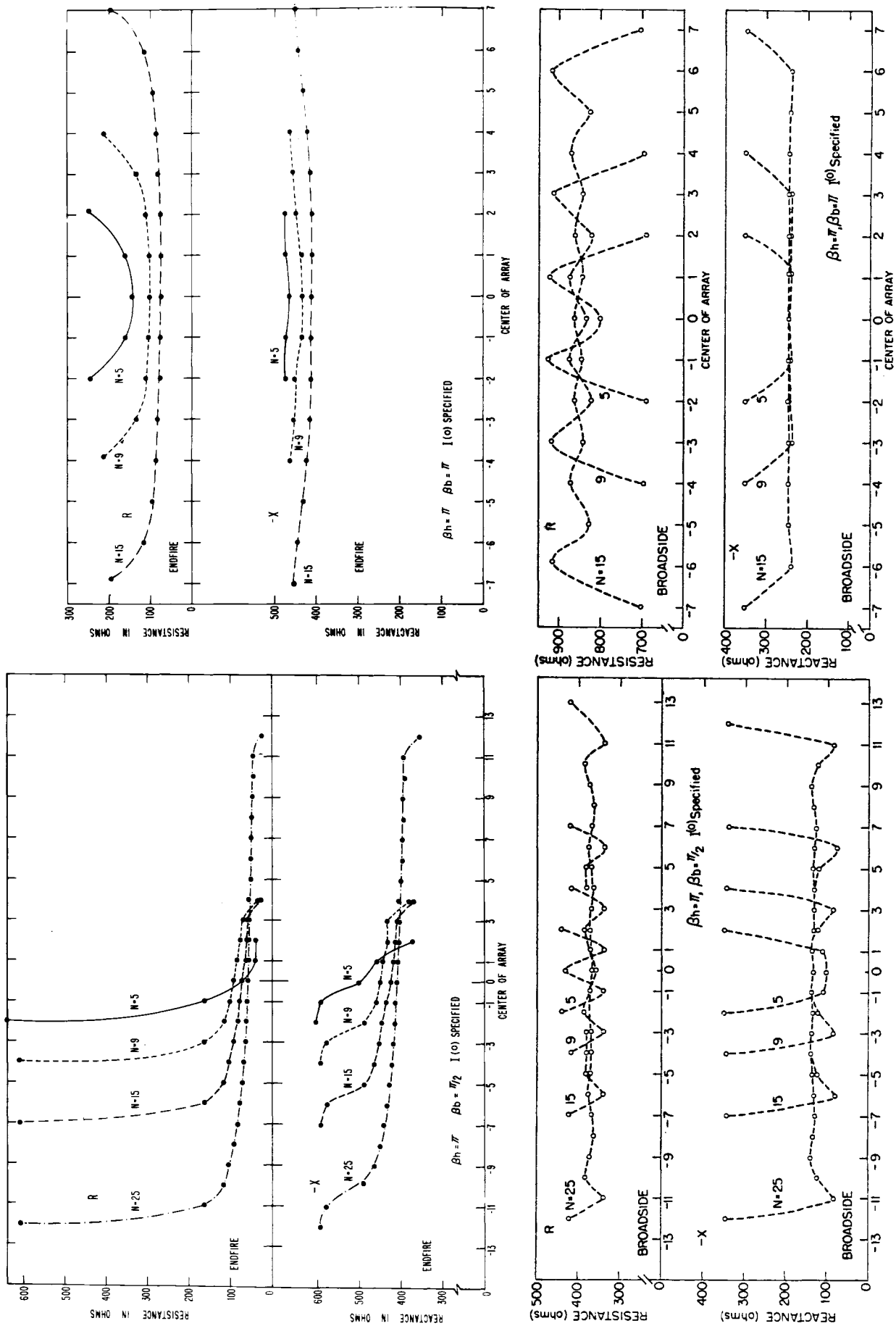

Fig. 11. Resistance and reactance of N-element curtain array of full-wave elements with spacing b. Upper left: unilateral endfire; upper right: bilateral endfire. Lower figures: broadside with two different spacings. Two-term theory.

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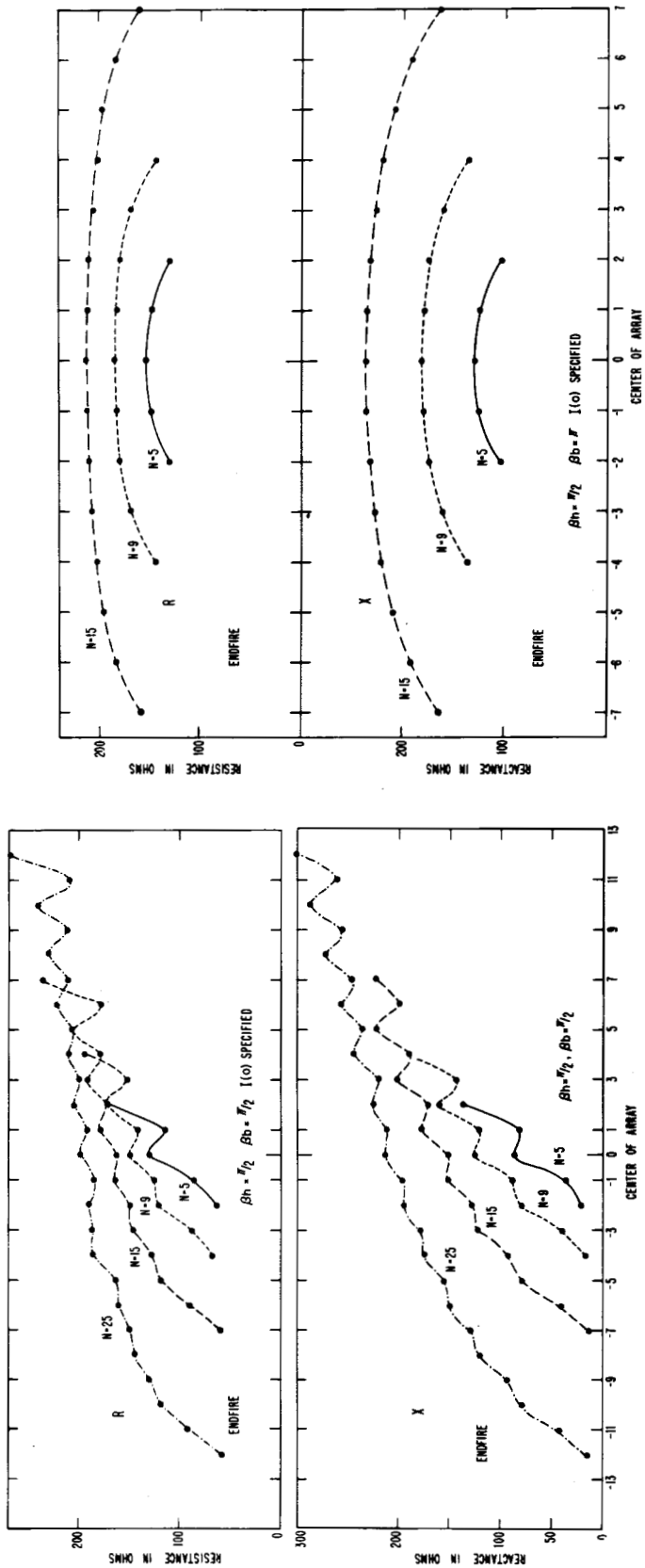

Fig. 12. Like upper figures in Fig. 11 but for half-wave instead of full-wave elements. Two-term theory.

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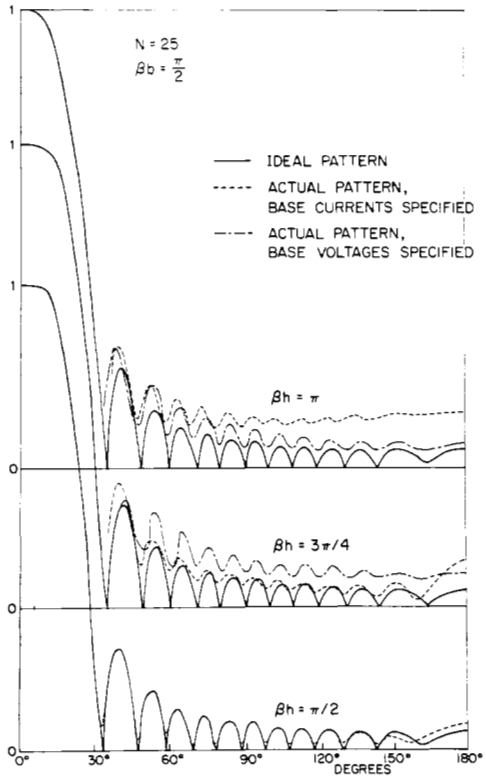

Fig. 13. Field patterns of unilateral end-fire arrays of 25 elements of length h and spacing b. In the ideal patterns (solid line) sinusoidal currents are assumed. The dotted lines show patterns determined by the two-term theory with current at the driving points specified; the dot-dash lines with driving voltages specified.

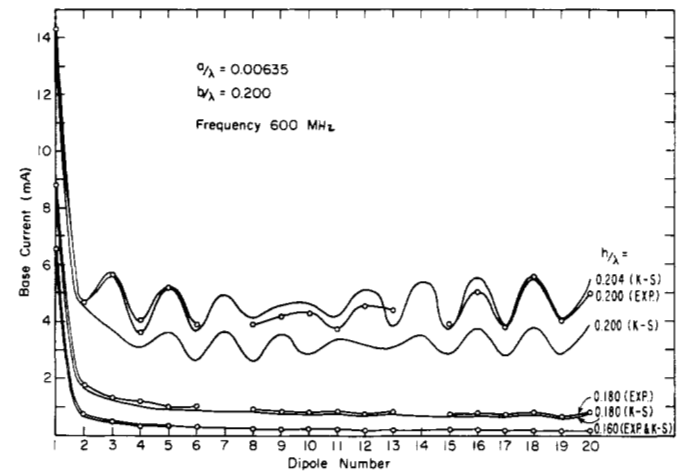

Fig. 14. Currents at the driving points of the cylindrical elements in a 20-element array of antennas of length h and radius a over a ground plane. Element 1 is driven, all others are parasitic. Solid lines: two-term theory; experimental points by Mailloux [45].

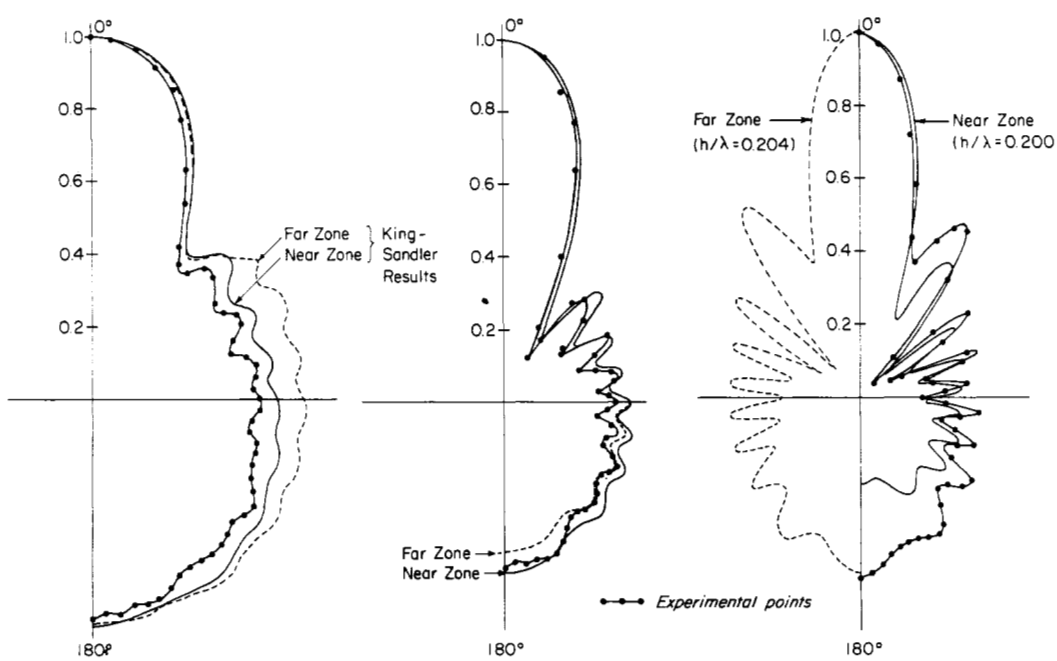

Fig. 15. Field patterns of the 20-element array. Solid and broken-line theoretical curves are two-term theory; experimental points by Mailloux [45].

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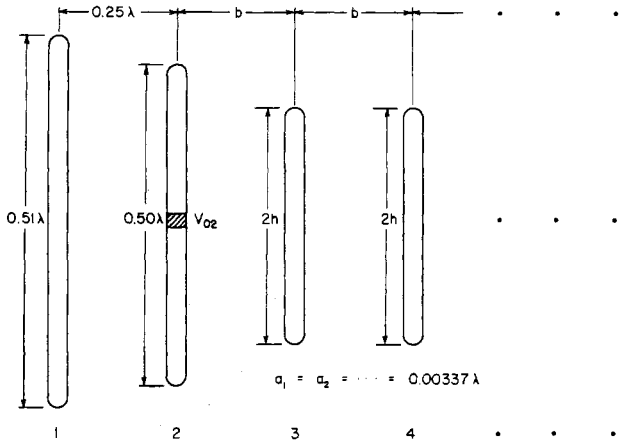

Fig. 16. A Yagi array with directors of length $2h$ and spacing b.

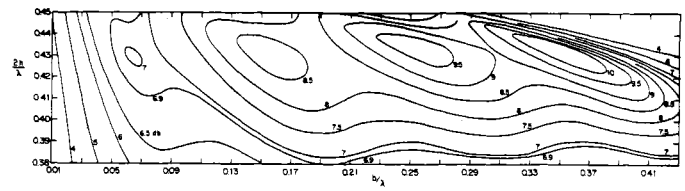

Fig. 17. Forward gain for a four-director array as in Fig. 16. Three-term theory.

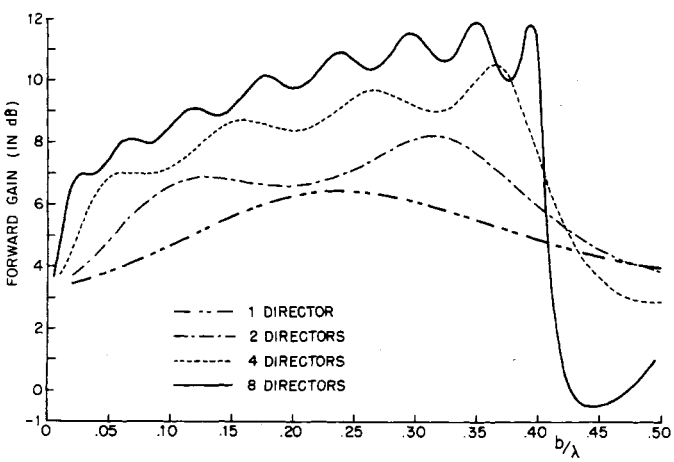

Fig. 18. Forward gain for a Yagi array with directors of length $2h \doteq 0.43\lambda$, radius $a = 0.00337\lambda$, and spacing b.

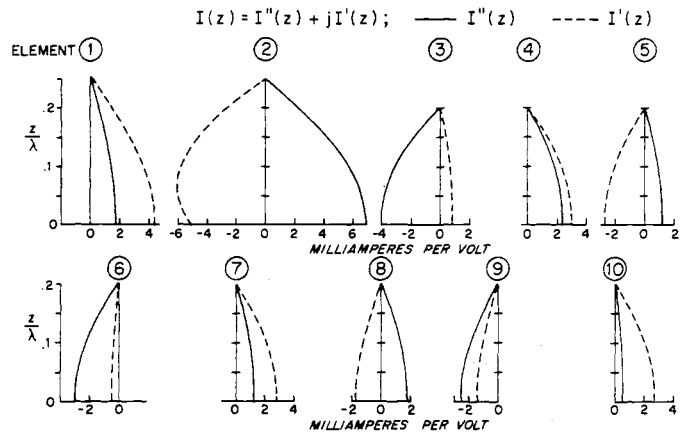

Fig. 19. Currents in a ten-element Yagi array. $I'(z)$ is in phase quadrature, $I''(z)$ is in phase with the driving voltage of the single driven element no. 2. $h_1 = 0.255\lambda$, $h_2 = 0.25\lambda$, $h_3 = \cdots = h_{10} = 0.20\lambda$. $b_{12} = 0.25\lambda$, $B_{23} + \cdots = b_{9,10} = 0.3\lambda \cdot = 2 \ln (2h/a) \doteq 10$ for $h \doteq 0.25\lambda$. Three-term theory.

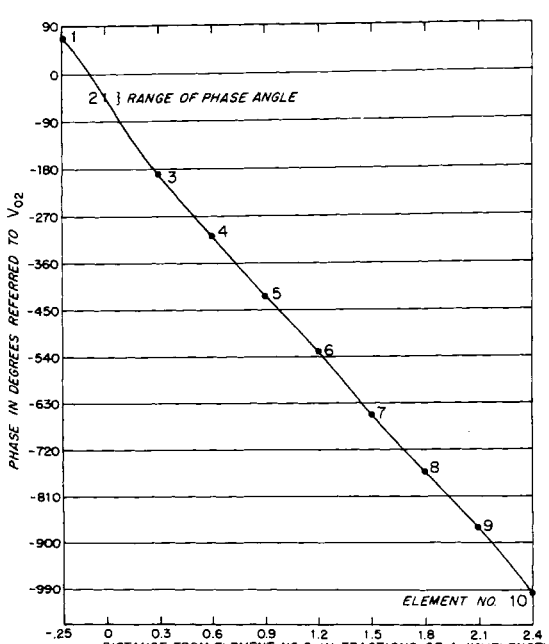

Fig. 20. Phases of the currents in the elements shown in Fig. 18.

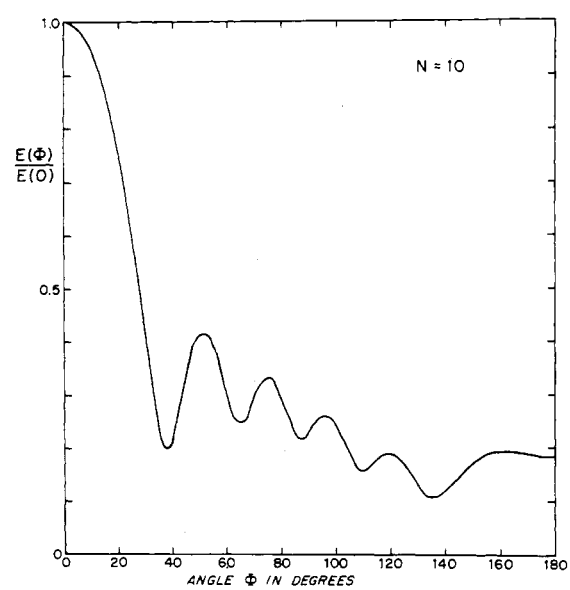

Fig. 21. Horizontal field pattern of the ten-element array with currents shown in Fig. 19.

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Electrically Thick Antennas and Arrays

The linearized integral equations (14) and (22) and the numerous approximate solutions that have been devised for them apply specifically to electrically thin antennas with $\beta a \leq 0.1$. For such antennas it is immaterial whether the exact kernel for the tube or the simpler approximate form [both are given in (2)] is used. The fact that the delta-function generator defined in (13) excites currents on both the inside and outside surfaces of the tube is also of no consequence, since the former are negligible when βa is small. When the restriction of the thickness of the antenna is removed and tubular antennas with βa as large as 2 are considered, an even more complicated problem emerges. In practice, antennas of such electrically large cross-sectional size (e.g., internally excited sections of missiles, millimeter-wave monopoles, and broadband monopoles) must be driven from an internally activated radial transmission line, or half of the dipole is driven over a conducting plane as the extension of the inner conductor of a thick coaxial line. In both cases the driving fields are maintained only on the outside surface of the tubular antenna. The inside is excited only by way of the open ends of the tube. This means that the correct boundary conditions to replace (13) for the rotationally symmetrical delta-function approximation of the actual driving mechanism are $E_z(a^+, z) = -V_0\zeta(z)$, $E_z(a_i^-, z) = 0$, $-h \leq z \leq h$, where a^+ is the outer radius, a_i^- the inner radius of the tube. It now turns out that the tubular antenna has become a doubly-connected region that requires a more general treatment. In particular, the integral equation (14) must be supplemented by an additional integral over the cross section of the tube at its center [50]. Moreover, the properties of the kernel which are utilized in approximate analytical solutions of the integral equation for thin antennas are no longer available, so that numerical methods must be used with the exact kernel. As a further complication, the possibility of transverse currents on the surface of the thick antenna cannot be overlooked. These can be investigated in terms of a non-rotationally symmetrical delta-function generator [51]. Thus, the tools are available for the study of isolated and coupled electrically thick antennas. Another approach is through circular arrays of more and more closely spaced elements.

Conclusion

Great advances have been made in the fundamental understanding of the properties of isolated and coupled cylindrical antennas, both in air and in dissipative media. No longer is it necessary to ignore the physical dimensions of the antennas and assume hypothetical distributions of current that take no account of the actual physical and electrical environment. The bridge built by Hertz from the mathematical theory of Maxwell to practical radio transmission has been extended. New and better tools are available for the antenna engineer.

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A Review of Ionospheric F Region Theory

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***Abstract*—Following a brief historical introduction, the paper reviews the types of data which concern ionospheric F region theory. The physical processes which are believed to control the variations of electron density in the F1 and F2 layers are then discussed. Possible explanations of several F region phenomena are presented, together with a brief consideration of magnetic storm effects.**

I. Introduction

A. The F Region

IONOSPHERIC PHYSICS has a standing of over forty years as an experimental subject. Its origins, however, can be traced back for over a century. The existence of an electrically conducting layer in the upper atmosphere was advanced by Gauss in 1839 and Kelvin in 1860, as a possible explanation of the magnetic daily variations, and formed part of Balfour Stewart's atmospheric dynamo theory of 1882 [1]–[3]. Following Marconi's transatlantic radio transmission in 1901, the presence of a reflecting layer was suggested independently by Kenelly and Heaviside in 1902. That this layer might consist of ionization produced by solar ultraviolet radiation seems to have been suggested by Taylor in 1903 and Fleming in 1906 [4]. Hence the concept of the ionosphere developed only a few years after Thomson's discovery of the electron, although the actual name did not appear in the literature until about thirty years later [5].

The scientific study of the ionosphere really began with the experimental determination of the height of the reflecting layers in 1925 by Breit and Tuve [6] and Appleton and Barnett [7]. Later the stratified nature of the ionosphere became apparent and naturally this raised the question of the physical origin of the various layers, a question that has taken decades to resolve. Even now many interesting problems remain, though much progress has been made since an important series of review articles appeared in these PROCEEDINGS in February 1959.

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